

FaceX-Zoo: A PyTorch Toolbox for Face Recognition

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ABSTRACT

Due to the remarkable progress in recent years, deep face recognition is in great need of public support for practical model production and further exploration. The demands are in three folds, including 1) modular training scheme, 2) standard and automatic evaluation, and 3) groundwork of deployment. To meet these demands, we present a novel open-source project, named FaceX-Zoo, which is constructed with modular and scalable design, and oriented to the academic and industrial community of face-related analysis. FaceX-Zoo provides 1) the training module with various choices of backbone and supervisory head; 2) the evaluation module that enables standard and automatic test on most popular benchmarks; 3) the module of simple yet fully functional face SDK for the validation and primary application of end-to-end face recognition; 4) the additional module that integrates a group of useful tools. Based on these easy-to-use modules, FaceX-Zoo can help the community to easily build state-of-the-art solutions for deep face recognition and, such like the newly-emerged challenge of masked face recognition caused by the worldwide COVID-19 pandemic. Besides, FaceX-Zoo can be easily upgraded and scaled up along with further exploration in face related fields. The source codes and models have been released and received over 900 stars at <https://github.com/JDAI-CV/FaceX-Zoo>.

CCS CONCEPTS

• **Computing methodologies** → **Biometrics**; *Image representations*; • **Software and its engineering** → *Software libraries and repositories*.

KEYWORDS

Face Recognition, Deep Learning, PyTorch Toolbox

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1 INTRODUCTION

In face recognition community, there emerges a number of excellent open-source projects facilitating the research and development for

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	FaceX-Zoo	InsightFace	face.evoLve	OpenFace
Modular&Scalable	✓			
SDK	✓			
Distr. Training	✓	✓		
Model Lightning	✓	✓	✓	
Mixed Precision	✓			
Masked Face	Data+ Algo.	Data		
License	Apache	MIT	MIT	Apache
Framework	PyTorch	MXNet	PyTorch	Torch

Table 1: Comparison of popular open-source projects for deep face recognition. *Modular&Scalable* indicates the intentional design with highly modular and scalable fashion. *SDK* indicates face SDK for end-to-end face recognition. *Distr. Training* indicates the availability of distributed training. *Masked Face* is the capability for masked face recognition.

deep face recognition. For example, InsightFace [1] is a classic toolbox for deep face analysis based on MXNet. face.evoLve [2] provides a comprehensive library for face related analysis. OpenFace [3] is a face library for the support of mobile device applications. Beyond that, the rapid development of deep face recognition pursues a significant need of a more comprehensive framework and standard evaluation to facilitate the research and development. To this end, we build a new toolbox, named FaceX-Zoo, in the form of PyTorch [14] library, which is highly modular, flexible and scalable (Table 1). It includes the state-of-the-art training pipeline for discriminative face feature learning, the standard evaluation towards fair performance comparison, and the SDK for efficient proof of concept and further applications of end-to-end face recognition. All the source codes and trained models are released to facilitate the community to develop their own solutions against various real-world challenges from the perspective of training, evaluation, and deployment. We hope that FaceX-Zoo is able to provide helpful support to the community and promote the development of face recognition.

2 CHARACTERISTICS

Modular and scalable design. FaceX-Zoo consists of four modules with respective functions. The modules and their sub-modules are developed in a decoupled style, following the principle of object-oriented design, so that usability and scalability are highly promoted. For example, one can easily set the choice of training mode, backbone, supervisory head and data sampler during the usage of FaceX-Zoo, by only editing the configuration file; researchers and developers can create and add new instances in these elements (training mode, backbone, etc.) in the training module, as well as many test protocols in the evaluation module, without modifying the core code. Moreover, FaceX-Zoo provides the face SDK and the

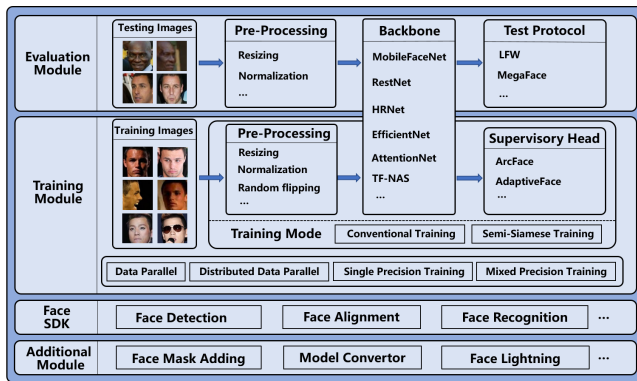


Figure 1: The architecture of FaceX-Zoo. The training and evaluation modules are specialized for face recognition, while the face SDK module empowers the end-to-end deployment of face detection, alignment and recognition. The additional module integrates a group of useful tools.

Backbone	MobileFaceNet, AttentionNet, ResNet, SE-ResNet, EfficientNet, GhostNet, TF-NAS, ResNeSt, ReXNet
Head	AM-Softmax, ArcFace, AdaCos, AdaM-Softmax, CircleLoss, CurricularFace, MV-Softmax, NPCFace, MagFace
Test	LFW, CPLFW, CALFW, AgeDB30, RFW, MegaFace, MegaFace-Mask

Table 2: Top and middle: state-of-the-art candidates of backbone and supervisory head implemented in FaceX-Zoo. Bottom: popular test sets applied in Test Protocol. The complete references of them can be found on the project page ¹.

additional modules for the flexible extension according to various demands.

State-of-the-art training. FaceX-Zoo provides state-of-the-art practices for face recognition model training, such as the advanced loss functions, special training modes, various backbone networks and data augmentation strategies for discriminative face feature learning.

Standardized evaluation module. There are a large number of evaluation benchmarks for face recognition, which require a unified implementation for efficient and fair evaluation. For example, the official test protocol of MegaFace [11] is implemented as a binary file, resulting in inconvenient test in many conditions. To address this issue, FaceX-Zoo provides a standard and open-source implementation for evaluation on many benchmarks with two mainstream protocols, called LFW-based and MegaFace-based. Users can evaluate their models and make fair comparison on various benchmarks by editing the configuration file efficiently.

Easy to use. The configuration is shaped by the ‘.yaml’ file, thereby one can conveniently customize the training and testing by editing the configuration file. The converting tool is provided to convert the trained model to ONNX or other required formats, so that one can easily deploy the models for application. Besides, we provide a simple yet fully functional face SDK written in python, facilitating the usage of provided modules and development of new functions.

¹<https://github.com/JDAI-CV/FaceX-Zoo/tree/main/reference>

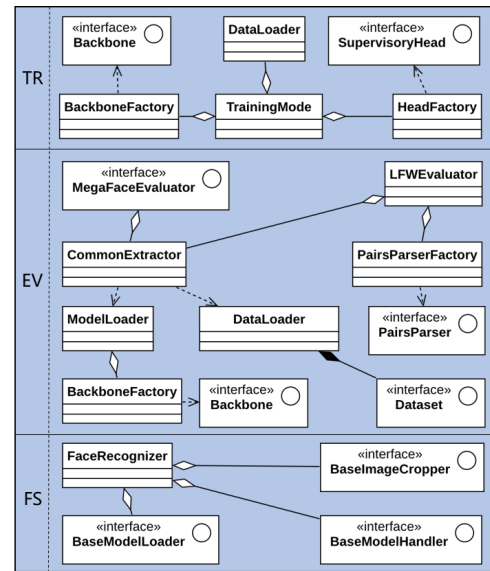


Figure 2: The class diagram of FaceX-Zoo. TR: training module. EV: evaluation module. FS: face SDK module. This design pattern is further reused in the additional module.

Support to new tasks. Recently, due to the pandemic of COVID-19, masked face recognition has attracted increasing attention in the face community. To develop the specialized model for masked face recognition, three essential components are indispensable: masked face training data, training algorithm and evaluation benchmark. As far as we know, FaceX-Zoo is the first project that provides all three components, bringing the advantage of deep face recognition from regular condition to masked face.

3 ARCHITECTURE

The overall architecture of FaceX-Zoo is presented in Fig. 1. The major class diagram is shown in Fig. 2. The framework consists of four parts: the training module, evaluation module, face SDK module and additional module. We elaborate on them as below.

3.1 Training Module

The training module is dedicated to the training of the face recognition network. It consists of the sub-modules of Pre-Processing, Backbone, Supervisory Head and Training Mode, accordingly implemented by the classes of DataLoader, BackboneFactory, HeadFactory and TrainingMode.

Pre-Processing. It fulfills the basic transformations on images before sending them to the network, including resizing, normalization, random cropping, random flipping, random rotation, *etc.* One can add the customized operations flexibly.

Backbone. FaceX-Zoo includes a series of state-of-the-art backbone architectures, which are listed in Table 2. Besides, any off-the-shelf choices can be easily customized in this project.

Supervisory Head. Supervisory head is defined as the supervision formulation and its corresponding computation module towards accurate face recognition. In order to learn discriminative features for face recognition, the predicted logits are usually processed by some

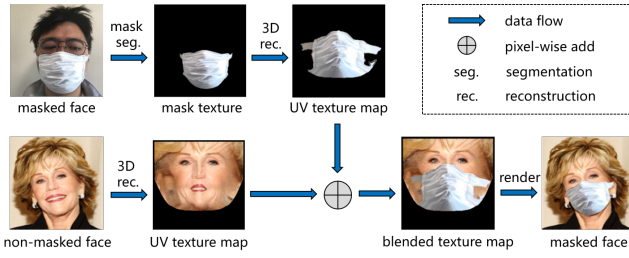


Figure 3: FMA-3D wears virtual mask on face image. The mask template is automatically segmented from arbitrary masked face images, and aligned on the UV texture map.

specific operations, such as normalization, scaling, adding margin, etc., before sending to the softmax layer. FaceX-Zoo implements a series of softmax-style losses listed in Table 2.

Training Mode. The conventional training mode of face recognition is regarded as the baseline routine. Concretely, it schedules the training inputs by DataLoader, then sends the inputs to the backbone for forward passing, and finally computes a criterion as the training loss for backward updating. In addition, We consider a practical situation in face recognition that is to train the network with shallow distributed data [6]. Accordingly, a recent promising training strategy is integrated to facilitate the training on shallow face data.

Implementation. As shown in Fig. 2, there are four major classes implemented to establish the training module. Specifically, TrainingMode is the core class to aggregate the other ones, including DataLoader, BackboneFactory and HeadFactory.

3.2 Evaluation Module

The evaluation module has most parts overlapped with the training, yet the incremental sub-module is Test Protocol.

Test Protocol. There are many benchmarks to measure the accuracy of face recognition models from various aspects. Many of them focus on specific face recognition challenges, such as cross age, cross pose, and cross race (Table 2). Among them, the most commonly used test protocols are based on the benchmarks of LFW [10] and MegaFace [11]. We integrate these protocols into FaceX-Zoo with simple usage and clear instruction, by which people can easily test their models on single or multiple benchmarks via simple configuration. Besides, it is convenient to extend additional test protocols in this module. It is worth noting that, based on MegaFace, we shape a masked face recognition benchmark with the same statistics of MegaFace, named MegaFace-Mask, which is dedicated to the evaluation of masked face recognition.

Implementation. LFWEvaluator and MegaFaceEvaluator are the core classes for the protocols of LFW and MegaFace, respectively. CommonExtractor is used for feature extraction, ModelLoader for loading the models, PairsParseFactory for parsing the test pairs, CommonMegaFaceEvaluator and MaskedMegafaceEvaluator for the test on regular and masked MegaFace, respectively.

3.3 Face SDK Module

The end-to-end face recognition pipeline includes not only face recognition, but also face detection and alignment. In order to

Backbone / Head	LFW	CPLFW	CALFW	AgeDB	MegaFace
MobileFaceNet [4]	99.57	83.33	93.82	95.97	90.39
ResNet152-irse [8]	99.85	89.72	95.56	98.13	97.48
EfficientNet-B0 [16]	99.55	84.72	94.37	96.63	91.38
TF-NAS-A [9]	99.75	85.90	94.87	97.23	94.42
Attention-56 [18]	99.88	89.18	95.65	98.12	97.75
AM-Softmax [17]	99.58	83.63	93.93	95.85	88.92
ArcFace [5]	99.57	83.68	93.98	96.23	88.39
MV-Softmax [19]	99.57	83.33	93.82	95.97	90.39
CircleLoss [15]	99.57	83.42	94.00	95.73	88.75
MagFace [13]	99.53	84.32	94.03	95.82	89.85

Table 3: The performance (%) of regular face recognition with different settings of backbone and supervisory head. Upper half: the supervisory head is set as MV-Softmx; lower half: the backbone is set as MobileFaceNet.

validate the effectiveness of trained models conveniently in the context of an end-to-end pipeline, FaceX-Zoo provides a simple yet fully functional module, named face SDK. It has two advantages. First, with the simple configuration, one can update the models in the pipeline without any change to the original code. Second, face SDK provides low-level interfaces to facilitate further development for more complex applications.

Implementation. Face SDK includes three core classes, named ModelLoader, ImageCropper and ModelHandler. They are in charge of loading models, cropping and normalizing facial area, and inference, respectively.

3.4 Additional Module

This module provides the useful tools which are commonly needed in practical model production. The first tool is for virtual mask adding, named FMA-3D (**3D-based Face Mask Adding**), which empowers to automatically augment a large amount of masked face data for training and testing masked face recognition model. As shown in Fig. 3, the photo-realistic masked face images are synthesized via 3D reconstruction and mask segmentation techniques [7, 12]. The second, *i.e.*, **Model Converter**, is for converting the PyTorch models to ONNX or other required formats, widening the application of produced models. The third, *i.e.*, **Face Lightning**, is for producing lightweight models by knowledge distillation techniques.

4 APPLICATION EXAMPLES

4.1 Regular Face Recognition

To facilitate reproducing state-of-the-art networks and fulfilling new explorations with FaceX-Zoo, we conduct a number of experiments with various choices of backbone and supervisory head for regular face recognition. The test results are partially shown in Table 3. One can refer to the page of *Training Mode*² for more details of training setting and complete test results. Here, Table 3 provides the key results by the state-of-the-art training towards regular face recognition.

²https://github.com/JDAI-CV/FaceX-Zoo/tree/main/training_mode

Training Mode	LFW	CPLFW	CALFW	AgeDB
Conventional Training	91.77	61.56	76.52	73.90
Semi-siamese Training	99.38	82.53	91.78	93.60

Table 4: The performance (%) of the training modes applied on shallow data, where only 2 images available per ID.

4.2 Shallow Face Learning

In many real-world scenarios of face recognition, the dataset is limited in depth, *i.e.*, only two face images are available for each ID. This task, so-called shallow face learning, is problematic to the conventional training methods for face recognition. FaceX-Zoo integrates the solution of Semi-Siamese Training [6], specialized for Shallow Face Learning, which has many differences from the conventional training mode. For a quick validation, we conduct a shallow face experiment with FaceX-Zoo and show the test results in Table 4, from which we can observe great advantage of the new training mode in FaceX-Zoo regarding the shallow face task.

4.3 Masked Face Recognition

Due to the recent worldwide pandemic of COVID-19, face recognition encounters the challenge of facial mask occlusion. FaceX-Zoo provides the solution of masked face recognition in three aspects. First, a new method named FMA-3D is proposed and implemented, which is dedicated to sampling and adding facial mask via 3D reconstruction and mask segmentation techniques (Fig. 3). Second, a large number of photo-realistic masked face images are synthesized by FMA-3D, for not only training data augmentation, but also test protocol mimicking the real-world masked face task. Third, a special modeling is designed with facial decomposition and model fusion strategy, of which the sub-networks are well trained and released in FaceX-Zoo. From the test comparison in Fig. 4, we can observe that the synthesized data and assembled model greatly improve the performance of masked face recognition.

5 CONCLUSION AND FUTURE WORK

FaceX-Zoo is a highly modular and scalable open-source framework for deep face recognition. It consists of four major modules: training, evaluation, SDK, and additional module. The training module enables training state-of-the-art face recognition networks with various choices of backbone, supervisory head, and training mode; the evaluation module provides the standard benchmark for fair and convenient testing; the SDK module completes the pipeline of end-to-end face recognition in the application; lastly, the additional module provides a number of useful tools, including facial mask adding, model convertor, and face lightning. These modules are designed in a scalable fashion, and constitute the framework of FaceX-Zoo which can be easily installed and utilized by the community. One can use FaceX-Zoo for practical model production, and also play with it as a prototype for further development, owing to the scalability.

In the future, FaceX-Zoo will be continually strengthened from three aspects. First, it will include new elements, such as face parsing and face attribute recognition, to thereby enrich the functionality “X” of FaceX-Zoo. Second, new backbone architectures and

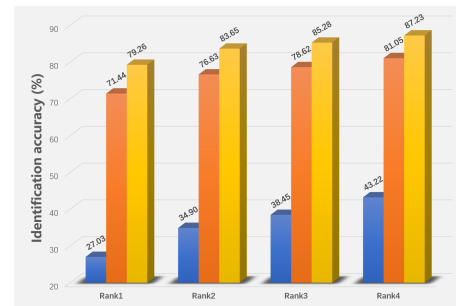


Figure 4: Rank-K identification accuracy on MegaFace-Mask. Best viewed in color. Blue: baseline. Orange: sub-networks. Yellow: masked face augmentation + sub-networks.

supervisory heads will be supplemented along with the development of deep learning techniques. Third, based on the state-of-the-art training towards discriminative visual features, FaceX-Zoo will widen its application on more tasks, such as face parsing, face attribute recognition, image retrieval *etc.*

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